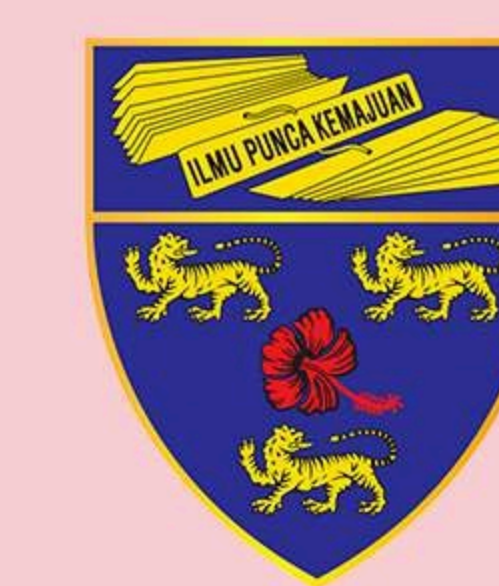


The Silent Exit: Predicting Customer Churn in the Malaysian Telecom Sector

by Goh Wee Ken



Problem Statement

Easily switch providers in Malaysia's "low-switching cost" market.

Early identification of at-risk customers is complex and challenging.

The Telecom Landscape in Malaysia

20%~40% Slice lost of customer

4x to 8x Expensive to retain existing customer.

Objective

To **develop** a predictive model that identifies customers likely to **churn**, to reduce attrition and protect revenue.



Proposed Dataset [1]

Datasets
MTD September 2019 & 2022 Customer Record

Total Rows
7,776 customers

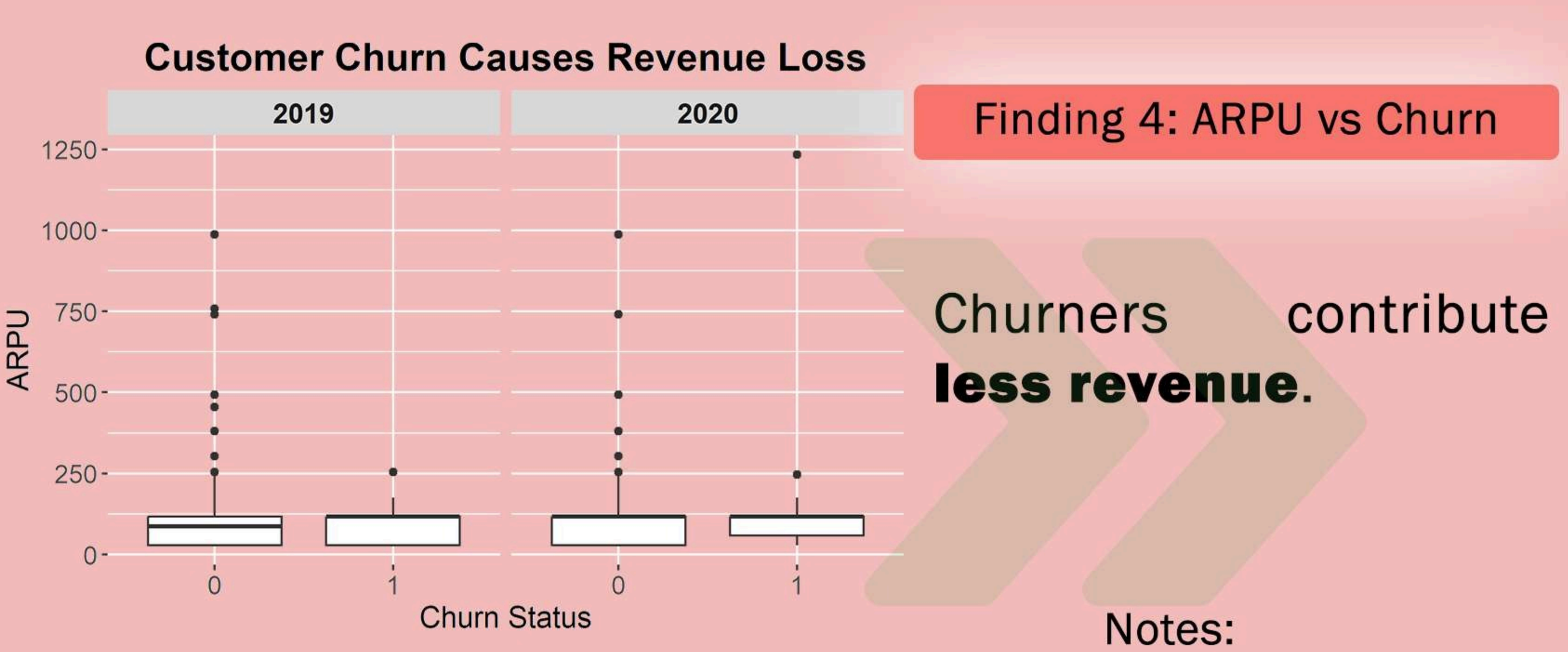
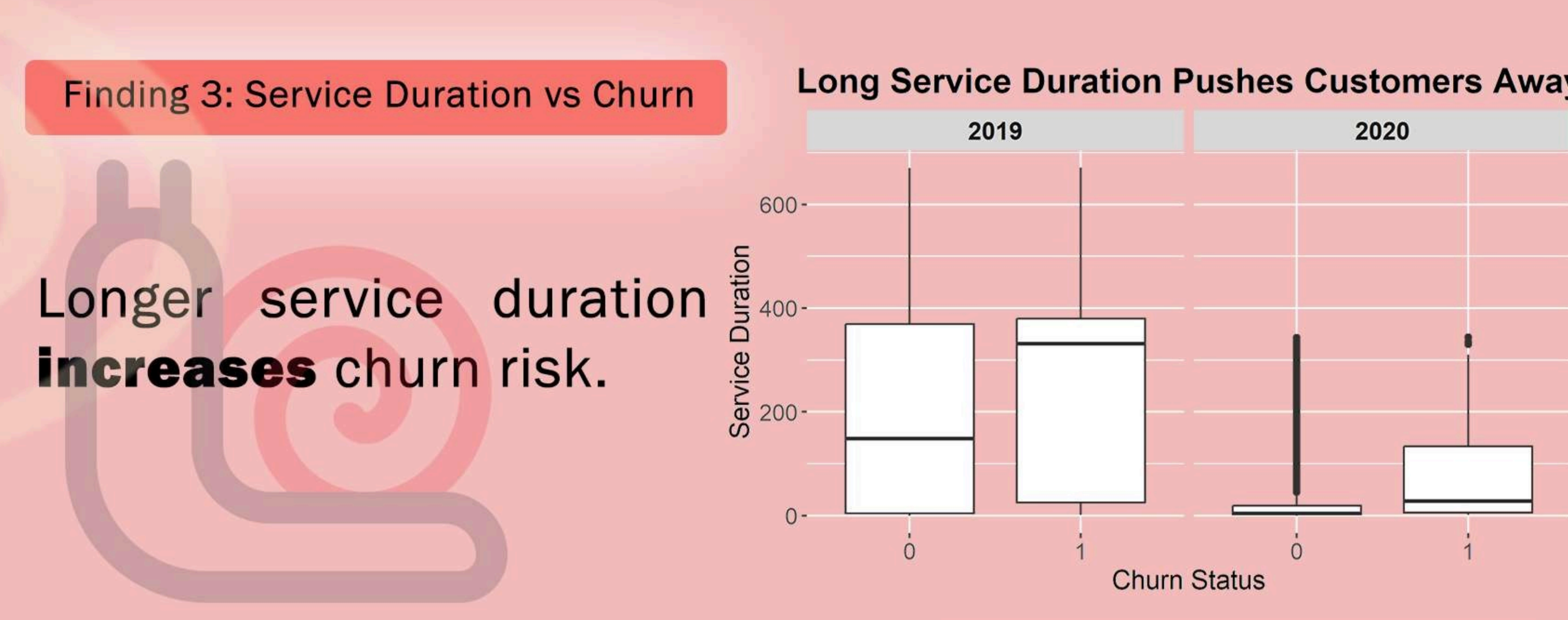
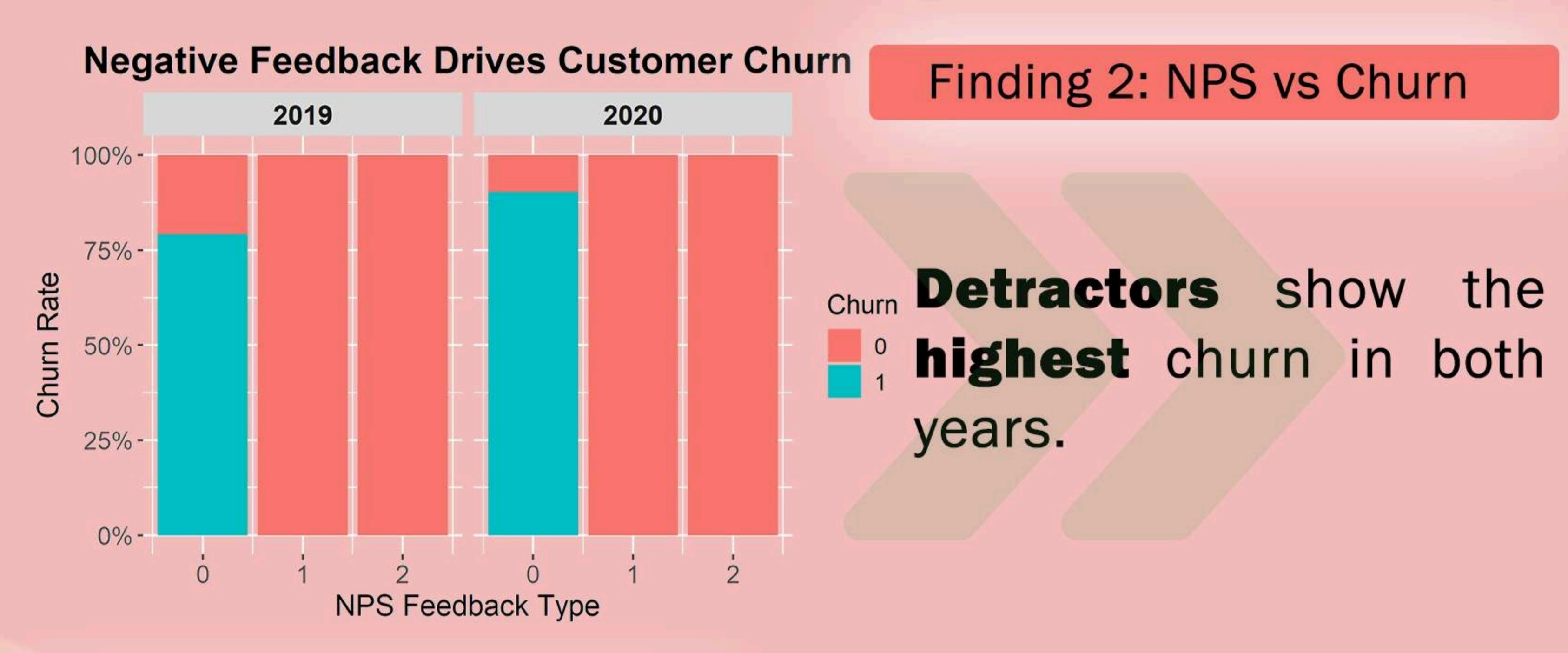
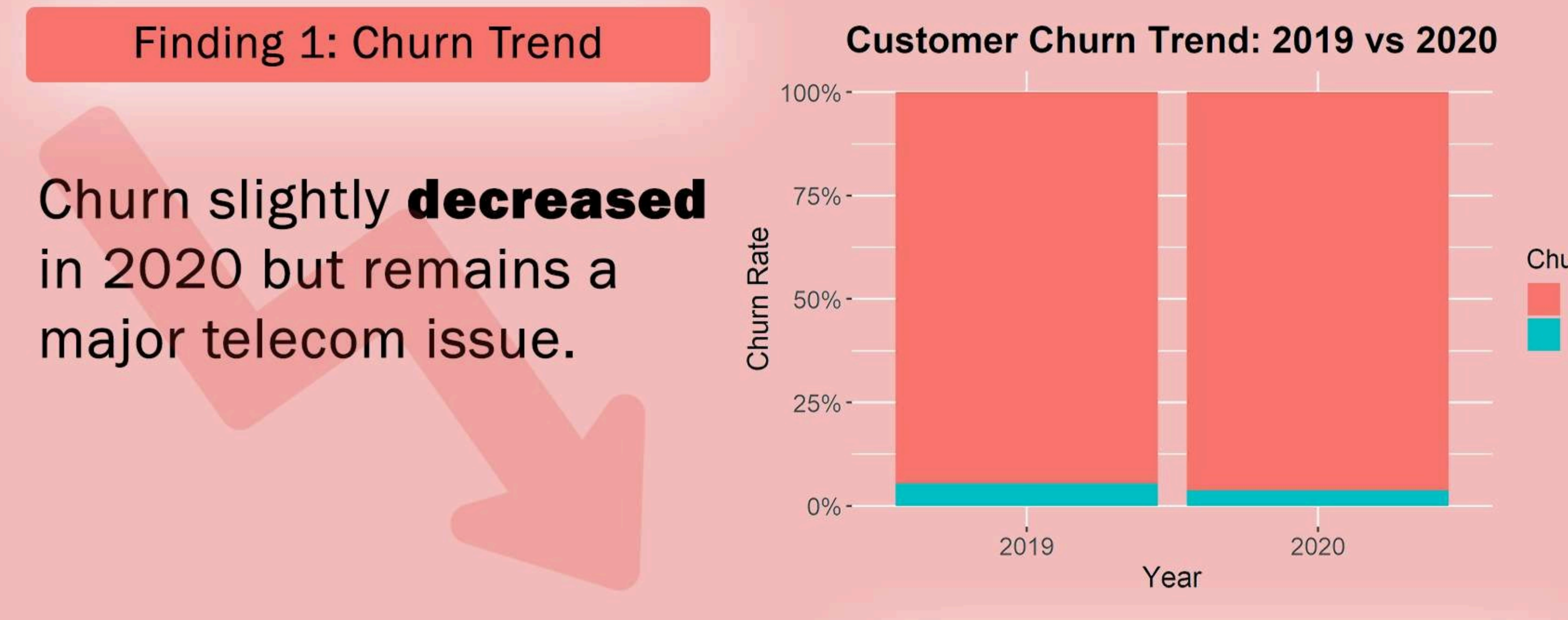
Variables
25 customer behavior and service variables.
Target Variables
POTENTIAL CHURNER

Model Logic



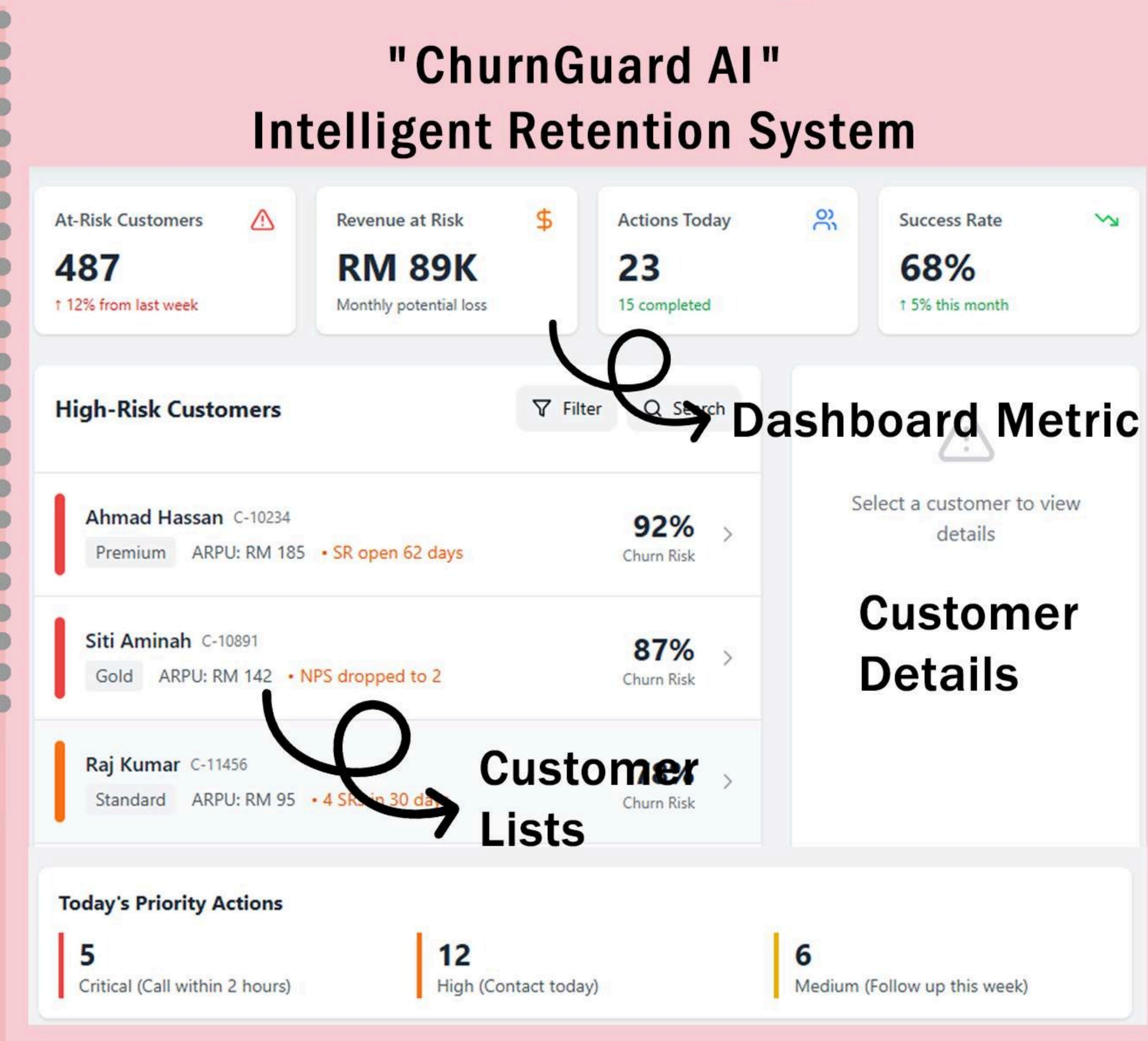
Exploratory Data Analysis (EDA)

What drives the customers away ?



Package used: ggplot2, dplyr, scales

Mockup Design



Features

- Dashboard Metrics**
Summary cards displaying critical business metrics.
- Customer Details Panel**
Selected customer's profile, risk factors, and recommended actions.
- Final Output**
 - ✓ Predicts customer churn
 - ✓ Early intervention
- Priority Action Bar**
Showing the number of customers requiring action.



References

[1] Mustafa, N., Ling, L. S., & Razak, S. F. A. (2021). Customer churn prediction for telecommunication industry: A Malaysian Case Study. F1000Research, 10, 1274. <https://doi.org/10.12688/f1000research.73597.1>